IT05: Robotics and Automatic Control ロボット工学と自動制御

Quiz 2: Concept of feedback control フィードバック制御の概念

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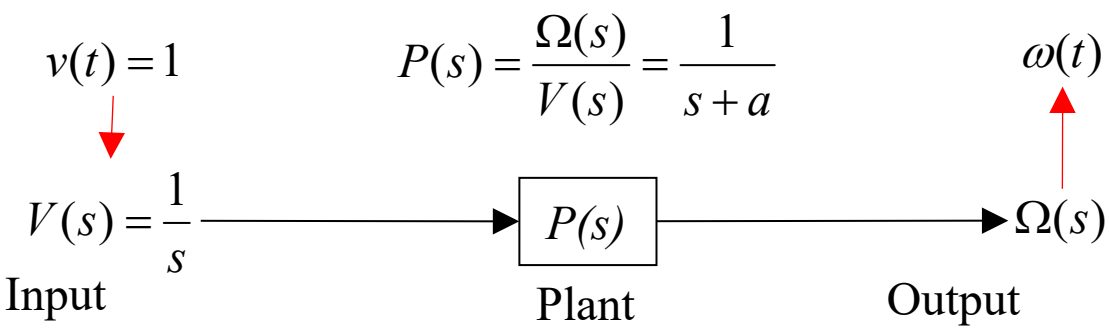
Derive a time domain output of the following system by your hand calculation, and understand the inverse Laplace transform better. You can write your derivation down on a sheet of paper, take a photo of it, paste it to this document, and submit the document. (10 points).

下のシステムの時間領域の出力を手計算で導出し，逆ラプラス変換について理解を深めよ．導出を紙に書き，その写真をとり，この文書にそれを貼り付けて提出しても良い．（10点）．



$$\cancel{\frac{-\frac{1}{a}}{s+a} + \frac{\frac{1}{a}}{s}} \qquad \cancel{\frac{s^2}{a} + \frac{s+a}{a}}$$

$$\frac{1}{s+a} \Rightarrow e^{-at}, \quad \frac{1}{Ts+1} = \frac{1}{T} \cdot \frac{1}{s+\frac{1}{T}} \Rightarrow \frac{1}{T} e^{-\frac{t}{T}}$$

$$\Omega(s) = \frac{K_e}{(T_m s+1)(T_e s+1)s} = \frac{A}{T_m s+1} + \frac{B}{T_e s+1} + \frac{C}{s}$$

$$\Rightarrow \omega(t) = \frac{A}{T_m} e^{-\frac{t}{T_m}} + \frac{B}{T_e} e^{-\frac{t}{T_e}} + C$$

then, $$\frac{K_e}{(T_m s+1)(T_e s+1)s} = \frac{As(T_e s+1) + Bs(T_m s+1) + C(T_m s+1)(T_e s+1)}{(T_m s+1)(T_e s+1)s}$$

$$= \frac{(AT_e + BT_m + CT_m T_e)^2 + (A+B+CT_m+CT_e)s + C}{(T_m s+1)(T_e s+1)s}$$

so, $$\begin{cases} AT_e + BT_m + CT_m T_e = 0 \\ A + B + CT_m + CT_e = 0 \\ C = K_e \end{cases}$$

$$A = -\frac{K_e T_m^2}{T_m - T_e}, \quad B = \frac{K_e T_e^2}{T_m - T_e}, \quad C = K_e$$

$$\omega(t) = -\frac{K_e T_m}{T_m - T_e} e^{-\frac{t}{T_m}} + \frac{K_e T_e}{T_m - T_e} e^{-\frac{t}{T_e}} + K_e$$

$$\omega(t) = K_e\left(1 - \frac{T_m}{T_m - T_e} e^{-\frac{t}{T_m}} + \frac{T_e}{T_m - T_e} e^{-\frac{t}{T_e}}\right)$$